2. Intensity Transformations

Using the original grayscale image:

a) Apply a **log transformation**

$$s = c \log(1 + r)$$

and display the result.

b) Apply gamma transformations using:

- γ = 0.4
- γ = 1.0
- γ = 2.5

Display the original and all three gamma-transformed images together.

c) Apply a **piecewise linear transformation** using:

$$(r_1, s_1) = (70, 20)$$
$$(r_2, s_2) = (180, 230)$$

d) Apply **automatic contrast stretching** so that the minimum intensity becomes 0 and the maximum intensity becomes 255.

Display all results with suitable titles.

3. Histogram Processing

Using the low-contrast grayscale image:

a) Calculate its **256-bin histogram** using NumPy and display the histogram.

b) Identify and print the intensity value that occurs most frequently.

c) Perform **histogram equalization manually using the cumulative distribution function (CDF)**.

d) Perform histogram equalization using:

```
cv2.equalizeHist()
```

e) Display:

- Original image
- Manually equalized image
- OpenCV equalized image

f) Plot the histograms of the original and equalized images and compare their intensity distributions visually.

4. Image Smoothing

Create two degraded versions of the grayscale image:

- one containing **Gaussian noise**
- one containing **salt-and-pepper noise**

Then:

a) Apply a **5 × 5 mean filter** to the Gaussian-noise image.

b) Apply a **5 × 5 Gaussian filter** with σ = 1.5 to the Gaussian-noise image.

c) Apply a **5 × 5 median filter** to the salt-and-pepper-noise image.

d) For the salt-and-pepper image, compare the results of:

- Mean filtering
- Gaussian filtering
- Median filtering

Display the noisy and filtered images together.

5. Image Sharpening

Using the original grayscale image:

a) Apply the following **Laplacian kernel** manually:

$$\begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

and display the edge response.

b) Apply the **Sobel operator** in both the horizontal and vertical directions and calculate the gradient magnitude.

c) Apply the **Prewitt operator** in both the horizontal and vertical directions and calculate its gradient magnitude.

d) Sharpen the original image using:

$$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 5 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

e) Display the following together for comparison:

- Original
- Laplacian response
- Sobel magnitude
- [redacted]
- Sharpened image